

Server

JPM proprietary server CPU shipment model

In this note, we introduce a JPM proprietary server CPU shipment model and forecast total server CPU shipments to grow from 26mn units in 2025 to 68mn by 2028 (38% CAGR). We segment server CPU demand into three buckets: (1) AI headnode CPUs, (2) agentic-AI server CPUs, and (3) general-purpose server CPUs. With AI accelerator shipments projected to show a ~50% CAGR over FY25–28 based on our CoWoS estimates by the semi team ([link](#)), we see a step-function increase in AI headnode CPU demand, reinforced by a rising CPU-to-accelerator attachment trend. While general-purpose server CPUs should continue to show a steady ~5% CAGR, agentic-AI CPUs are the breakout growth vector, with 155% CAGR expected over 2025–28. Net-net, we believe overall general server shipments can sustain 20%+ growth in the coming years. This super-cycle is positive for TSMC, Unimicron, ASPEED, Lotes, Wiwynn, Tripod, GUC, ASE, and memory vendors, such as SK Hynix and Samsung Electronics, across Asia Technology.

- **Server CPU landscape reshaped by AI-led demand.** We forecast AI accelerator shipments to rise from 7.6mn units in 2024 to 32.5mn by 2028, with increasing headnode CPU attach rates due to more complex AI accelerators. The accelerator-to-CPU ratio is also compressing—from ~4:1 in legacy HGX servers to ~2:1 in NVL72 compute trays, and potentially ~1:1 in next-generation TPU designs. ARM-based CPUs are taking share, driven by Nvidia's Vera and Google's Axion, and we estimate ARM could represent ~90% of AI headnode CPU demand by 2028. We also expect ARM to gain share in agentic-AI servers, as hyperscalers lean into in-house CPU roadmaps for emerging workloads. As a result, we model ARM-based CPUs reaching ~43% of the blended server CPU market by 2028 (up from ~22% in 2025). That said, we still expect the x86 CPU market to see an accelerated 25% CAGR over the period (vs. flattish to LSD CAGR in the prior cycle).
- **New AI inference orchestration drives agentic-AI server demand.** As AI workloads shift from training towards inference, we expect hyperscalers to deploy more general-purpose servers around accelerators to improve Total Cost of Ownership (TCO). The incremental demand shows up primarily in external storage servers (e.g., KV cache) and compute servers that handle query orchestration around GPUs. While there is no fixed accelerator-to-CPU server ratio, our back-of-the-envelope analysis suggests ~0.5x (1 CPU for two accelerators) this year, with the ratio rising over time as accelerator complexity increases. This will become the key incremental driver of general server demand (>50% of the general server CPU market by 2028 vs. <10% last year).
- **Stocks leveraged to the super server CPU upcycle.** There is a high correlation between server CPU, BMC, substrate, and socket demand. We expect ASPEED, Lotes, and Unimicron to be the key beneficiaries of this super server CPU cycle, driven by strong unit demand, continued content upgrades, and potential product price hikes. AMD's continued m/s gain in x86 server CPUs also bodes well for TSMC and ASE as the key foundry/OSAT vendor. DRAM vendors could benefit from the increasing DIMM/SOCAMM demand. In the downstream, we believe the server PCB makers (e.g. Tripod) and general server ODMs (e.g. Wiwynn, Inventec) could also benefit from the strong server demand.

Technology - Hardware

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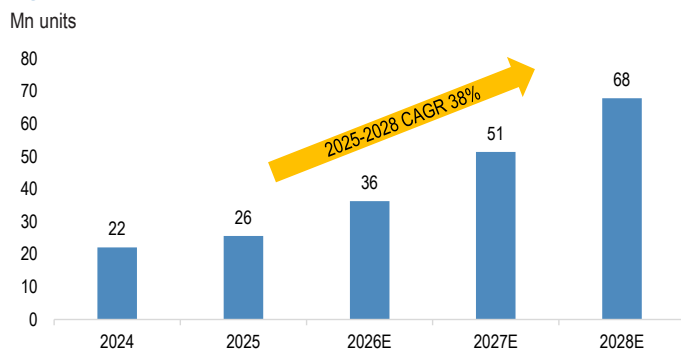
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Accelerating server CPU growth cycle on AI inference cycle

We introduce our proprietary server CPU demand model. We segment CPU demand into three buckets—(1) AI server headnode CPUs, (2) regular server CPUs, and (3) agentic-AI CPUs—and forecast total server CPU shipments to grow from 26mn units in 2025 to 68mn by 2028 (38% CAGR). As AI applications transition from training toward inference, we expect agentic-AI and AI headnode CPU demand to drive a “super” general server CPU cycle.

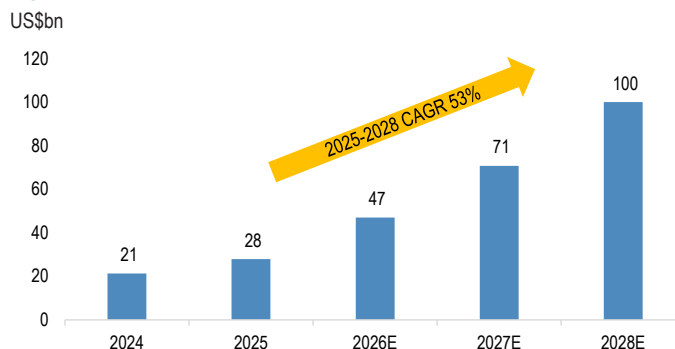
We also size the server CPU TAM by combining our shipment forecast with ASP assumptions across the three sub-segments. Given ongoing supply tightness and continued CPU upgrade cycles, we assume ~10% annual CPU price increases. As a result, we forecast server CPU TAM to show a 53% CAGR over 2025–28, reaching ~US\$100bn by 2028, driven by both volume growth and sustained ASP expansion.

Figure 1: Server CPU shipment trend and CAGR



Source: J.P. Morgan estimates. Gartner®.

Figure 2: Server CPU revenue TAM forecast and CAGR



Source: Gartner®, J.P. Morgan estimates.

Top-down methodology

We have anchored the total server CPU shipments for 2024 using Gartner data and forecast the server CPU demand for the respective segments over the coming years. We forecast AI server headnode CPU demand based on our AI accelerator forecast by the JPM semi team ([link](#)) (led by Gokul Hariharan) and the ratio of AI accelerators to headnode CPUs per server node. We assume no agentic AI activity in 2024 and derive the implied regular server CPU demand for 2024.

For the year 2025 and 2026, we assume non-AI server headnode CPU shipments (regular servers plus agentic AI CPUs) grow at 11%/30% YoY to 23mn/29mn, up from 20mn in 2024. This, coupled with 3mn/7mn of AI server headnode demand, could lead to 26mn/36mn total server CPU shipments (up 16%/42% YoY) in 2025/2026.

Beyond 2026, we anticipate 42%/32% YoY growth in total server CPU shipments, driven by rising attach rates of agentic AI CPUs relative to AI accelerators, strong AI server demand, and a 5% CAGR in regular server CPUs. Overall, we forecast a 38% CAGR in server CPU shipments through 2025-2028.

Table 1: Server CPU forecast - top-down methodology

Unit in millions	2024	2025	2026E	2027E	2028E	2025-28 CAGR
1. AI SERVER HEADNODE						
Global AI accelerator Shipments (mn)	7.6	9.6	16.1	25.2	32.5	50%
YoY		27%	67%	56%	29%	
Avg accelerators per CPU in headnodes	4.1	3.0	2.3	2.2	1.9	
Headnode Server CPUs Needed (mn)	1.8	3.2	7.1	11.3	16.8	74%
YoY		73%	124%	59%	48%	
2. REGULAR SERVER GROWTH (non AI workloads)						
Traditional server unit for non AI workload (mn)	11.6	11.9	12.5	13.1	13.8	
YoY	2%	3%	5%	5%	5%	
CPU per server	1.8	1.8	1.8	1.8	1.8	
non AI workload CPU (mn)	20.3	20.9	21.9	23.0	24.2	5%
YoY		3%	5%	5%	5%	
3. AGENTIC AI CPU CLUSTERS						
CPU per accelerator ratio	0.0	0.17	0.45	0.68	0.83	
TOTAL AGENTIC AI CPUs (mn)	0.0	1.6	7.3	17.1	27.0	155%
YoY			352%	134%	58%	
Total server CPU shipment (mn)						
Total server CPU shipment	22.1	25.7	36.3	51.4	67.9	38%
YoY		16%	42%	42%	32%	
Total server CPU shipment excl. AI server headnodes serve	20.3	22.5	29.2	40.1	51.1	31%
YoY		11%	30%	37%	27%	

Source: J.P. Morgan estimates. Gartner@data for 2024 total server CPU shipments. Global AI accelerator shipments are based on our CoWoS estimates by the semi team ([link](#)).

Strong accelerator demand + Rising CPU attach rate = Exponential AI server headnode CPU growth

We have seen a rising attach rate of AI server headnode CPUs. The number of AI accelerators per CPU in each AI server decreases from 4:1 to 2:1 and even 1:1 in the future generations, especially in AI ASIC servers. We attribute this to the rising requirement of CPU orchestration in AI servers due to the increasing complexity of AI server architectures. Of note, the AI accelerator forecasts are based on the CoWoS estimates by the JPM semi team ([link](#)) (led by Gokul Hariharan).

- **Nvidia GPU servers:** Accelerator to CPU ratio declines to 2:1 in NVL72 compute trays from 4:1 in HGX servers. We also see CPU demand in the NVLink switch tray of the NVL72 server rack. Interestingly, we see an increase in the accelerator to CPU ratio in the HGX servers in future generations (from 4:1 to 8:1), likely due to the relatively lower CPU requirements of the HGX server architecture.
- **AMD GPU servers:** Our research indicates 4:1 GPU to CPU ratios in the MI300 series and AMD MI455 Helio servers.
- **Google TPU servers:** Our research indicates that there are two designs for TPU v8 servers with 1:1 and 2:1 TPU-to-CPU ratios, and that there could be a 1:1 for future generations. We assume a 50%/50% mix for 1:1 and 2:1 design for TPU v8 generation.
- **AWS Trainium servers:** Our research indicates that the Trainium to CPU ratio will decline to 4:1 for the T3 liquid cooling project from 16:1 for the T2 and air-cooled T3 projects.
- **MSFT/Meta/other AI ASIC servers:** We assume 4:1 accelerators to CPU ratio for the rest of the AI ASIC servers.

The rising attach rate of AI server headnode CPUs, combined with strong AI accelerator demand, could drive exponential growth of AI server headnode CPU demand. Overall,

we forecast AI server headnode CPU to grow at a ~74% CAGR in 2025-2028.

Table 2: AI server headnode CPU forecast

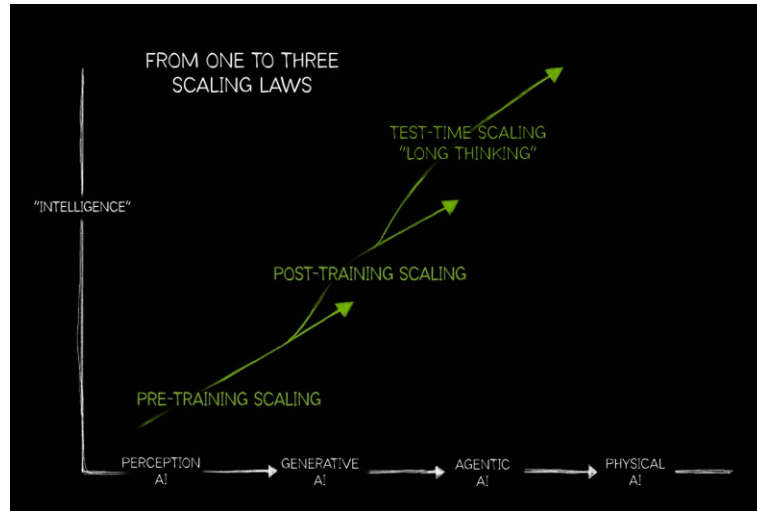
AI accelerator unit (mn)	2024	2025	2026E	2027E	2028E	Note
NVDA	4.5	6.2	8.9	10.0	11.6	
HGX (x86 base)	4.3	2.6	3.0	2.1	2.0	
NVL72 (ARM base)	0.1	3.6	5.9	7.9	9.6	
AMD	0.5	0.7	0.7	1.5	2.3	
TPU	1.7	1.8	4.4	8.9	11.6	
v4/5/6/7	1.7	1.7	2.7	1.7	-	
v8	-	0.1	1.7	7.2	8.6	
v9	-	-	-	-	3.0	
AWS	0.8	1.5	1.9	3.5	3.5	
Inferentia 2	0.5	0.1	-	-	-	
Trn 2	0.3	1.5	0.6	-	-	
Trn 3	-	-	1.3	3.5	-	
Trn 4	-	-	-	-	3.5	
Meta	-	-	0.3	0.5	1.1	
Microsoft	-	-	0.2	0.8	0.8	
Others	-	-	0.2	0.4	0.8	
Total AI accelerator units (mn)	7.4	10.1	16.5	25.6	31.7	
Avg number of accelerators per CPU	2024	2025	2026E	2027E	2028E	
NVDA	3.9	2.5	2.4	2.4	2.3	8 GPU to 2 CPU
HGX (x86 base)	4.0	4.0	4.0	8.0	8.0	2 GPU to 1 CPU; R200 changed to 8:1 GPU/CPU ratio
NVL72 (ARM base)	2.0	2.0	2.0	2.0	2.0	
AMD	4.0	4.0	4.0	4.0	4.0	
TPU	4.0	3.8	1.5	1.5	1.3	
v4/5/6/7	4.0	4.0	1.5	1.5	1.5	Assume v4/v5 4:1 & v6/7 2:1
v8	1.5	1.5	1.5	1.5	1.5	v8 has two versions, 1:1 or 2:1
v9	1.0	1.0	1.0	1.0	1.0	
AWS	5.3	14.5	14.3	10.0	2.0	Assume average 8 Inferentia to 2 CPUs
Inferentia 2	4.0	4.0	4.0	4.0	4.0	
Trn 2	16.0	16.0	16.0	16.0	16.0	
Trn 3	13.6	13.6	13.6	10.0	6.4	80% air cooling (16:1) and 20% liquid cooling (4:1), 50%/50% and 20%/80% AC/LC ratio in 2027/28
Trn 4	2.0	2.0	2.0	2.0	2.0	JPMc
Meta	4.0	4.0	4.0	4.0	4.0	
Microsoft	4.0	4.0	4.0	4.0	4.0	
Others	4.0	4.0	4.0	4.0	4.0	
Blended ratio	4.0	3.2	2.3	2.3	1.9	
AI headnode CPU units (mn)	2024	2025	2026E	2027E	2028E	
NVDA	1.1	2.4	3.7	4.2	5.1	
AMD	0.1	0.2	0.2	0.4	0.6	
TPU	0.4	0.5	2.9	5.9	8.7	
AWS	0.1	0.1	0.1	0.4	1.7	
Meta	-	-	0.1	0.1	0.3	
Microsoft	-	-	0.1	0.2	0.2	
Others	-	-	0.1	0.1	0.2	
Total	1.8	3.2	7.1	11.3	16.8	

Source: J.P. Morgan estimates. AI accelerator unit estimates are based on our CoWoS estimates by the semi team ([link](#)).

Agentic AI CPUs becoming a new driver for server CPU demand

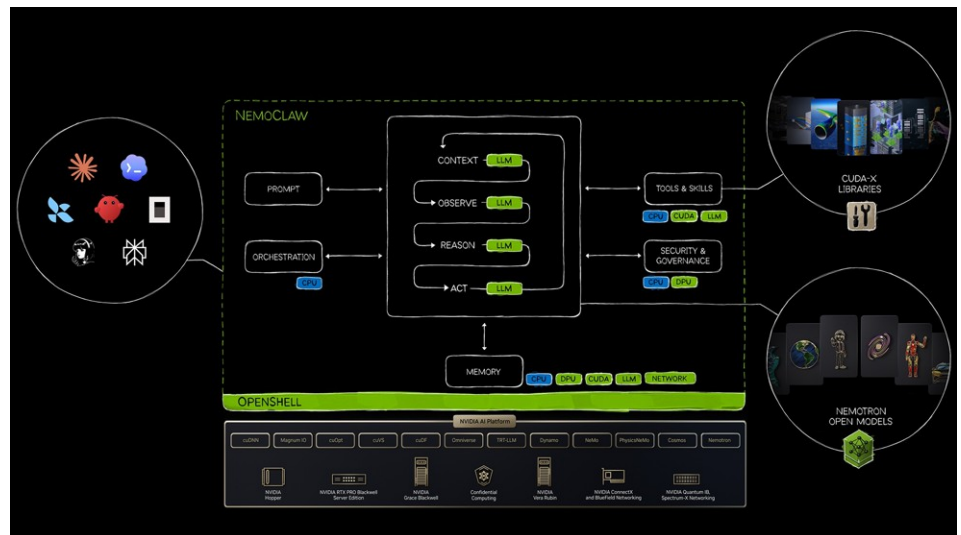
As we migrate towards the era of agentic AI, focusing on test-time scaling (i.e. long thinking), CPUs are playing an increasingly important role in orchestration, running tools and skills, storage, and security. For example, the Vera CPU not only sits in the NVL72 rack systems, but also serves as a standalone CPU across the entire Vera Rubin platform, including storage servers and CPU servers.

Figure 3: Stages of Intelligence



Source: Nvidia.

Figure 4: Agent = LLM + Harness; CPU plays a critical role in the agentic AI ecosystem



Source: Nvidia.

We attribute the strong general server demand this year (30-40% YoY growth on our estimates) primarily to the emerging demand for agentic AI. To quantify agentic AI demand, we assume no agentic AI activity in 2023/2024 and calculate the implied ratio of agentic AI CPUs to AI accelerators in 2025 and 2026. We then forecast the agentic AI CPU demand based on our estimates of this ratio and AI accelerator demand for 2027/2028.

To derive the implied ratio of agentic AI CPU demand to AI accelerator demand, we assume a ~3% CAGR for the regular server demand in 2023-2026 and derive the agentic AI CPU demand for 2025/2026 after subtracting the AI server headnode CPU and regular server CPU demand from our total server CPU demand. We expect this ratio

to rapidly increase in the coming years given the fast-growing agentic AI demand.

Overall, we forecast agentic AI CPU demand to show a 155% CAGR during 2025-2028, which we view as the key growth driver of total server CPU demand.

Triangulating top-down forecasts with bottom-up estimates

In the analysis below, we triangulate our top-down forecasts with bottom-up estimates. We anchor 2024 and 2025 Intel/AMD server CPU shipments to Gartner data. For 2025–28, we forecast x86/Arm server CPU shipments at 25%/72% CAGR, respectively. We estimate CPU demand for the remaining ASIC CPU vendors based on our industry research and our AI server headnode CPU demand framework.

Table 3: JPM server CPU bottom up estimates

Bottom up CPU unit breakdown	2024	2025	2026E	2027E	2028E	2025-28 CAGR
x86	20.6	19.9	25.8	33.2	38.8	25%
ARM	1.5	5.8	10.5	18.2	29.1	72%
Total	22.1	25.7	36.3	51.4	67.9	38%
% x86	93%	78%	71%	65%	57%	
% ARM	7%	22%	29%	35%	43%	

Source: Gartner®, J.P. Morgan estimates.

For each vendor, we break demand into three buckets: AI server headnode CPUs, regular server CPUs, and agentic-AI CPUs. We anchor AI headnode CPU demand using our AI server headnode CPU forecast exercise (Table 2). We assume only modest agentic-AI CPU deployments in 2024/2025 and derive the implied regular server CPU demand. For 2026–28, we assume a 5% CAGR for regular server CPU demand across categories.

We then use agentic-AI CPU demand (2026–28) and the remaining CPU buckets as balancing items to reconcile our bottom-up build with the top-down totals. Overall, we expect agentic-AI CPU demand to rise across the board.

Table 4: JPM server CPU bottom-up estimates - breakdown by applications

Bottom Up - CPU split by applications (mn units)	2024	2025	2026E	2027E	2028E
x86	20.6	19.9	25.8	33.2	38.8
General	18.9	17.9	18.8	19.8	20.7
Agentic AI	-	0.6	4.5	11.3	16.4
AI headnodes	1.7	1.3	2.5	2.2	1.6
Arm-based	1.5	5.8	10.5	18.2	29.1
General	1.4	2.9	3.1	3.3	3.4
Agentic AI	0.0	1.0	2.8	5.8	10.6
AI headnodes	0.1	1.8	4.6	9.1	15.1
Total server CPU demand	22.1	25.7	36.3	51.4	67.9

Source: Gartner®, J.P. Morgan estimates.

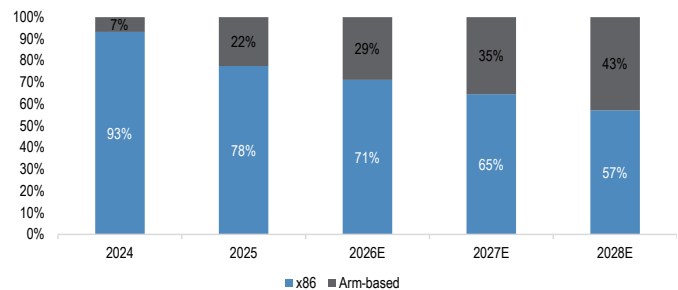
Faster ramp of Arm-based CPUs driven by rising Arm-based CPU attach rate

We forecast the adoption rate of Arm-based CPUs to reach ~43% by 2028 vs. ~22% in 2025, driven by a rising adoption rate in AI server headnodes and an x86 CPU shortage. We estimate that Arm-based CPUs will account for 90% of total AI server headnode CPU demand by 2028 vs. <10% in 2024, mainly driven by the ramp of the NVL72

server architecture with the adoption of Nvidia’s Arm-based CPUs and Google’s adoption of its in-house CPU for its TPU servers.

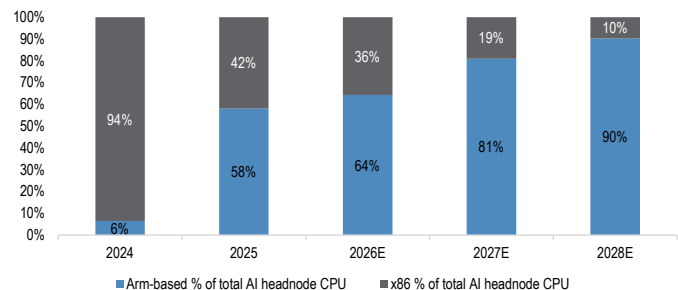
Unlike traditional servers, which host high core-count CPUs with better multi-threading and core-partitioned capabilities, AI server headnodes and agentic AI servers require lower latency and power-efficient CPUs to process AI workloads. Arm-based CPUs, such as Nvidia’s Vera, could be purpose-built for AI tasks. Overall, we expect Arm-based and x86 CPU demand to show a 72%/25% CAGR during 2025-2028, respectively.

Figure 5: Total server CPU - x86/ Arm-based shipment mix



Source: Gartner®, J.P. Morgan estimates. Other primarily consists of Arm-based CPUs.

Figure 6: AI server headnode - x86/ Arm-based shipment mix



Source: Gartner®, J.P. Morgan estimates.

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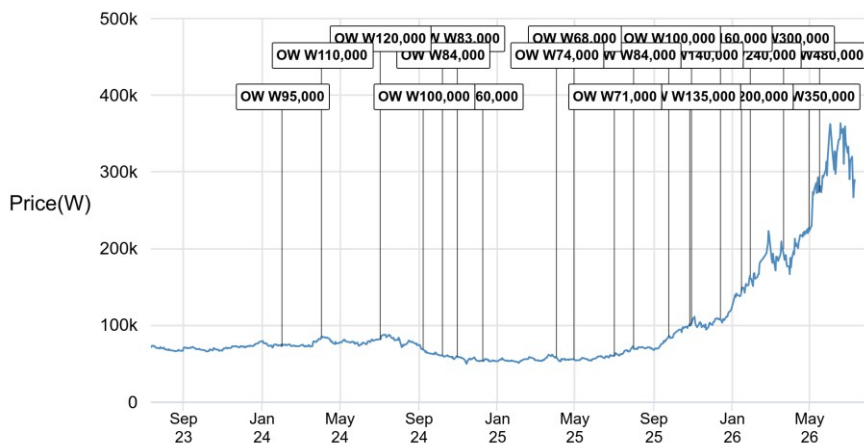
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Samsung Electronics (005930.KS, 005930 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Oct 09, 2011. All share prices are as of market close on the previous business day.

Date	Rating	Price (W)	Price Target (W)
01-Feb-24	OW	72700	95,000
03-Apr-24	OW	85000	110,000
03-Jul-24	OW	81800	120,000
08-Sep-24	OW	68900	100,000
08-Oct-24	OW	61000	84,000
31-Oct-24	OW	59100	83,000
10-Dec-24	N	53400	60,000
03-Apr-25	OW	58800	74,000
30-Apr-25	OW	55800	68,000
02-Jul-25	OW	60200	71,000
01-Aug-25	OW	71400	84,000
25-Sep-25	OW	85400	100,000
27-Oct-25	OW	98900	135,000
30-Oct-25	OW	101700	140,000
14-Dec-25	OW	108500	160,000
16-Jan-26	OW	145000	200,000
29-Jan-26	OW	164300	240,000
22-Mar-26	OW	199800	300,000
30-Apr-26	OW	226500	350,000
17-May-26	OW	273500	480,000

SK hynix (000660.KS, 000660 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Dec 31, 2000. All share prices are as of market close on the previous business day.

Date	Rating	Price (W)	Price Target (W)
07-Nov-23	OW	133000	160,000
25-Jan-24	OW	141500	175,000
22-Feb-24	OW	149000	185,000
28-Mar-24	OW	181200	260,000
16-Jul-24	OW	230000	300,000
08-Sep-24	OW	156400	240,000
24-Oct-24	OW	196000	260,000
10-Dec-24	OW	168900	210,000
23-Jan-25	OW	225500	280,000
03-Apr-25	OW	197900	300,000
24-Apr-25	OW	181000	285,000
25-Jun-25	OW	278500	360,000
24-Sep-25	OW	361000	460,000
23-Oct-25	OW	481000	650,000
29-Oct-25	OW	519000	700,000
14-Dec-25	OW	571000	800,000
16-Jan-26	OW	756000	1,000,000
29-Jan-26	OW	862000	1,250,000
22-Mar-26	OW	1006000	1,550,000
23-Apr-26	OW	1220000	1,800,000
17-May-26	OW	1848000	3,000,000

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.
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